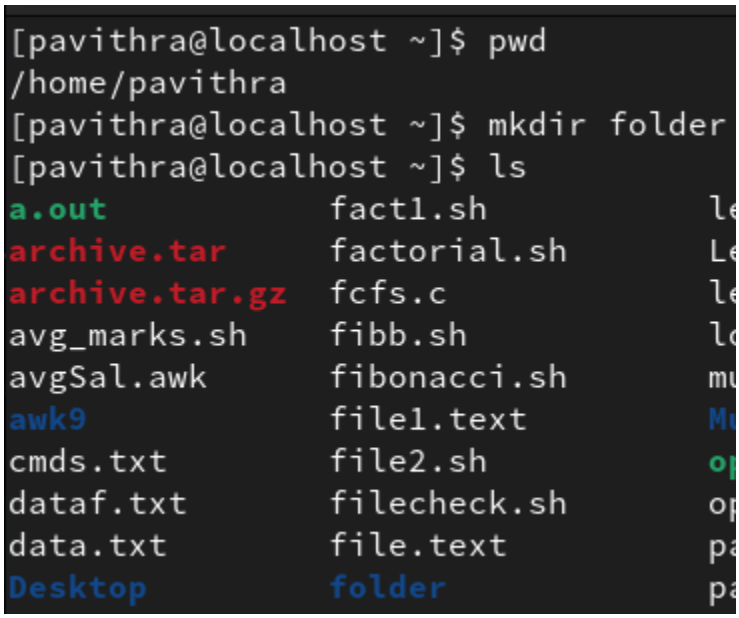
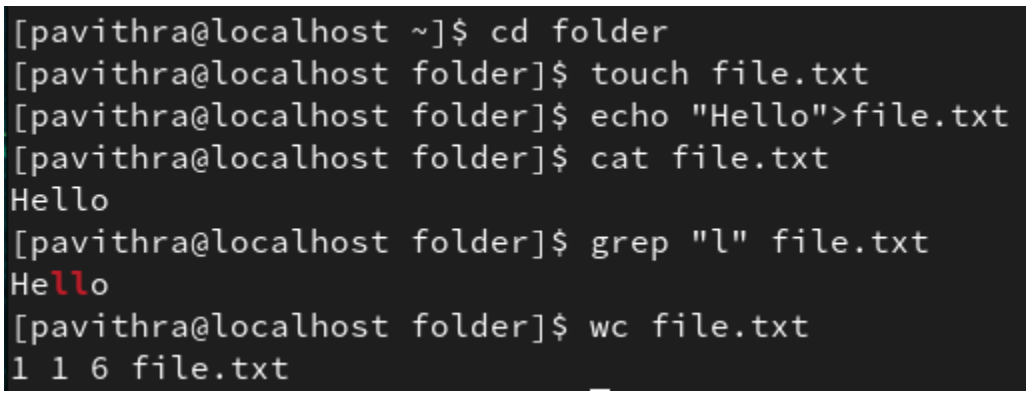
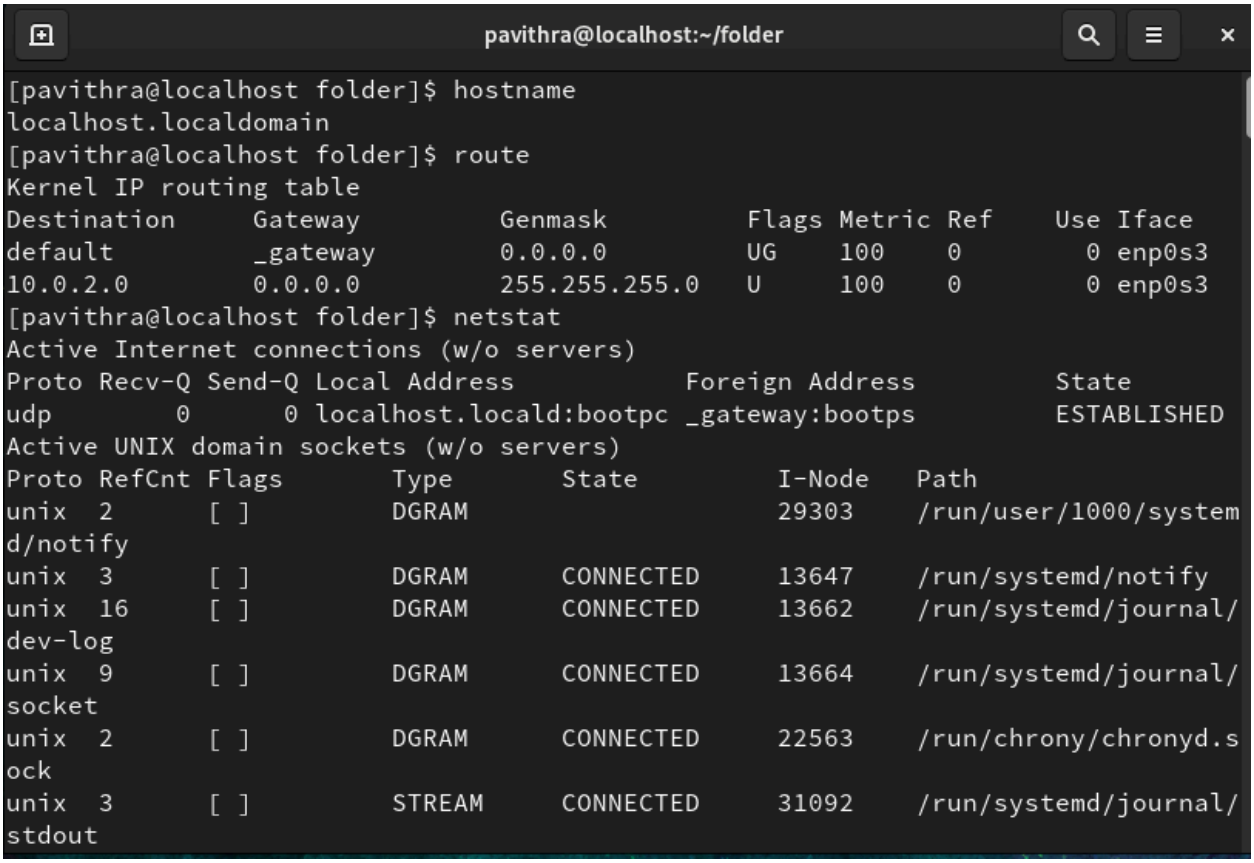
EXPERIMENT 1
BASIC LINUX AND NETWORKING COMMANDS

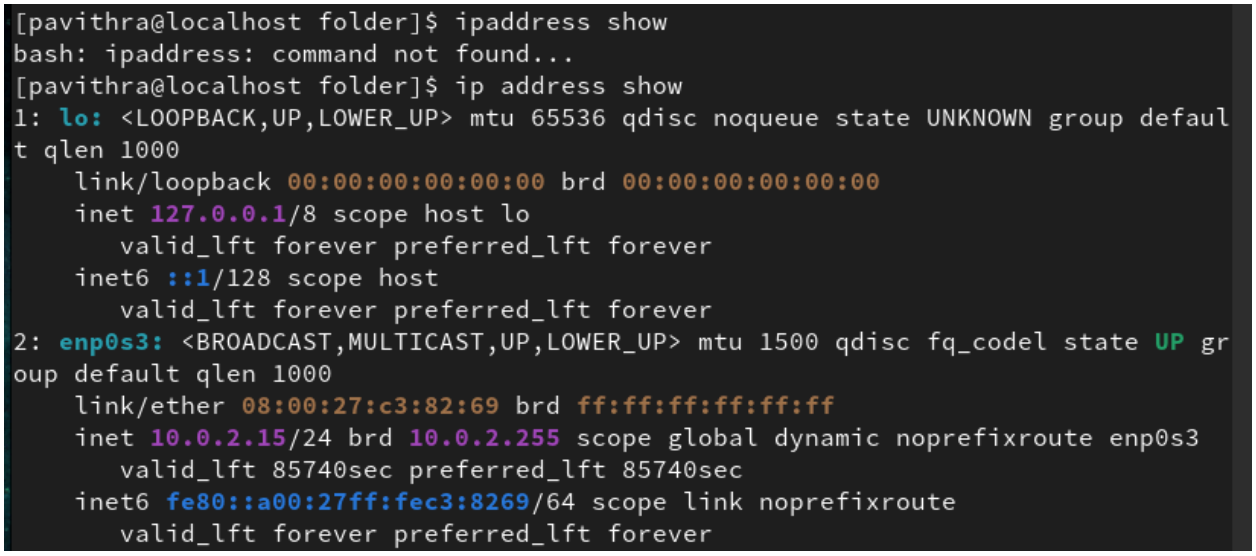
1. Basic Linux Commands:





2. Networking Commands:





```
[pavithra@localhost folder]$ nslookup www.google.com
Server:         192.168.88.1
Address:        192.168.88.1#53

Non-authoritative answer:
Name:   www.google.com
Address: 142.251.157.119
Name:   www.google.com
Address: 142.251.150.119
Name:   www.google.com
Address: 142.251.153.119
Name:   www.google.com
Address: 142.251.152.119
Name:   www.google.com
Address: 142.251.154.119
Name:   www.google.com
Address: 142.251.151.119
Name:   www.google.com
Address: 142.251.155.119
Name:   www.google.com
Address: 142.251.156.119
Name:   www.google.com
Address: 2001:4860:4827:7700::
```

```
[pavithra@localhost folder]$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
        inet6 fe80::a00:27ff:fec3:8269  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:c3:82:69  txqueuelen 1000  (Ethernet)
        RX packets 1054078  bytes 1567764419 (1.4 GiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 69002  bytes 4258978 (4.0 MiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 20  bytes 2190 (2.1 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 20  bytes 2190 (2.1 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

```
[pavithra@localhost folder]$ ping www.google.com
PING www.google.com (142.251.151.119) 56(84) bytes of data.
64 bytes from 142.251.151.119 (142.251.151.119): icmp_seq=1 ttl=117 time=8.95 ms
64 bytes from 142.251.151.119 (142.251.151.119): icmp_seq=2 ttl=117 time=5.33 ms
64 bytes from 142.251.151.119 (142.251.151.119): icmp_seq=3 ttl=117 time=6.01 ms
64 bytes from 142.251.151.119 (142.251.151.119): icmp_seq=4 ttl=117 time=5.04 ms
64 bytes from 142.251.151.119 (142.251.151.119): icmp_seq=5 ttl=117 time=5.13 ms
64 bytes from 142.251.151.119 (142.251.151.119): icmp_seq=6 ttl=117 time=4.64 ms
64 bytes from 142.251.151.119 (142.251.151.119): icmp_seq=7 ttl=117 time=5.78 ms
64 bytes from 142.251.151.119 (142.251.151.119): icmp_seq=8 ttl=117 time=8.99 ms
^C
--- www.google.com ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 7012ms
rtt min/avg/max/mdev = 4.642/6.234/8.990/1.629 ms
[pavithra@localhost folder]$ arp -a
_gateway (10.0.2.2) at 52:54:00:12:35:02 [ether] on enp0s3
```

```
[pavithra@localhost folder]$ traceroute www.google.com
traceroute to www.google.com (142.251.155.119), 30 hops max, 60 byte packets
 1  _gateway (10.0.2.2)  0.266 ms  0.191 ms  0.220 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * * *
13  * * *
14  * * *
15  * * *
16  * * *
17  * * *
18  * * *
19  * * *
20  * * *
21  * * *
22  * * *
```